# Twelve positive variables and materialized coefficients for the consistency of ZFC

Review and smaller constructions for Vladimir Reshetnikov

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**Abstract**

We review the new repository packages 09–15 and combine their primitive-root, signed-domain, ordinary-Pell and compiler ideas. The principal new polynomial has twelve strictly positive unknowns, exact degree 1257 and a fully charged 139-operation straight-line program. It eliminates four supplied witnesses at the price of one weighted primitive element. Applying rational Pell saturation before recovering the main quotient removes an unnecessary index restriction. The same idea yields a polynomial in 26 unrestricted integers of degree 2465 with 192 operations. Two earlier twelve-variable alternatives are retained with their complete proofs. Separate grouped equations improve the 15-variable degree-44 polynomial from 78 operations to 76.

A directed-signal Wang compiler with 943 tiles reduces the radix to $B = 2^{5^{10}}$. Its four actual integer coefficients are supplied as compressed binary payloads: 5,312,269 bytes uncompressed and 20,260 bytes in deterministic gzip form. This is not merely a recipe for inaccessible astronomical numerals. The mathematical reduction still depends on the imported native-window/Pell and exact K215 calibration theorems. We distinguish those dependencies, new mathematical proofs, finite circuit and coefficient checks, and the absent end-to-end proof-assistant certificate. No proof of consistency, accepting K215 execution, or global minimum is claimed.

## 1 Results, domains, and what is actually proved

The review is pinned to the immutable ProveIt revision

`f0ae4e0991ca44a7bb4e5671d42e33d775e48035`.

The previous review used revision `3b201cdea32d2c816f5f3d20734051f25fd3ef14`. The new tree replaces packages 01–08 by 09–15; package 15 is the previously delivered round-two release, not an independent new discovery. The source ledger records 127 fetched inert files, including their immutable URLs, Git blob identifiers, SHA-256 values and byte counts. All verified bytes match their Git blobs. No received repository program was run, no software was installed, and no repository was published or changed.

**Theorem 1** (Scope of the constructions)**.** *Fix the coherent four-coefficient tuple $(\beta, K_0, M_C, M_F)$ in Section 9. Relative to the imported K215 calibration and native-window/Pell compiler theorems identified below, the explicitly given new positive-domain polynomials have a solution in $\mathbb{Z}_{>0}$ exactly when the blank K215 machine halts, equivalently when a finite ZFC refutation exists. Thus ZFC proves that nonexistence of their positive solutions is equivalent to* Con(ZFC)*.*

*The 26-variable polynomial has the corresponding assertion over $\mathbb{Z}$. For a positive-domain construction, replacing each variable $v$ by $n_v+1$ gives a construction over $\mathbb{N} = \{0, 1, \ldots\}$ with unchanged variable count and degree. This translation is not asserted to preserve the reported operation count.*

This is an ordinary mathematical composition, intended to be formalizable in ZFC. It is not a newly checked Lean/Rocq theorem or a formal ZFC derivation supplied in the archive. In particular it does not prove Con(ZFC). Two-sided independence is not inferred from bare consistency alone.

| New presentation | Domain/variables | Equations | Operations | Degree |
|---|---|---|---|---|
| Saturated four-root product | $\mathbb{Z}_{>0}^{12}$ | 1 | 139 | 1257 |
| Its comparison system | $\mathbb{Z}_{>0}^{12}$ | 3 | 135 | 1248 |
| Restricted-$i$ four-root product | $\mathbb{Z}_{>0}^{12}$ | 1 | 140 | 1545 |
| Its comparison system | $\mathbb{Z}_{>0}^{12}$ | 3 | 136 | 1536 |
| Scaled-$y$ four-root product | $\mathbb{Z}_{>0}^{12}$ | 1 | 140 | 1625 |
| Its comparison system | $\mathbb{Z}_{>0}^{12}$ | 3 | 136 | 1616 |
| Signed saturated four-root product | $\mathbb{Z}^{26}$ | 1 | 192 | 2465 |
| Signed weighted three-root product | $\mathbb{Z}^{27}$ | 1 | 136 | 1116 |
| Signed gap-norm alternative | $\mathbb{Z}^{28}$ | 1 | 127 | 426 |
| Signed normalized two-root merger | $\mathbb{Z}^{28}$ | 1 | 128 | 258 |
| Grouped double-gap sum of squares | $\mathbb{Z}_{>0}^{15}$ | 1 | 76 | 44 |
| Grouped auxiliary-gap sum of squares | $\mathbb{Z}_{>0}^{15}$ | 1 | 75 | 50 |
| Grouped first-gap sum of squares | $\mathbb{Z}_{>0}^{15}$ | 1 | 73 | 70 |
| Grouped ordinary sum of squares | $\mathbb{Z}_{>0}^{15}$ | 1 | 72 | 80 |

Degree for a system means the maximum residual degree. One binary addition, subtraction or multiplication costs one operation, including a square or multiplication by a fixed numeral. Reusing an existing register and testing equality are free. The fixed coefficients are not supplied unknowns. This is an arithmetic-circuit metric, not bit complexity, expanded monomial count, or a measure of the size of a proof.

**Recommendation.** Use the degree-1257 twelve-variable product if the primary objective is supplied positive unknowns. The primary construction retains unrestricted positive $i$ and has immediate completeness from the old normalized witnesses. Its signed extension gives the 26-integer result. Use the degree-44 grouped polynomial when low degree matters more than unknown count. Coefficient height is independent of those tradeoffs: all explicitly certified compatible forms may use the entire Wang943 tuple. The previously known 67-operation polynomial and 32-variable quartic are not improved here.

## 2 What the recent commits add

The following is a comparison of representative pinned constructions, before the new work of this report. Metrics are variables / degree / operations unless stated otherwise.

| Package | Representative result | | Contribution and caveat |
|---|---|---|---|
| 09 combined13 | 13/571/84; 14/121/76 | 13/515/85; | Weighted merger of three roots, two-root merger, and a 32-variable quartic. The denominator proof is essential; a raw unweighted resultant is not enough. |
| 10 synthesis14 | 15/56/74 | | Groups unit equations before summing squares. The norm factors must first exclude their negative-unit branches. |
| 11 T4 | 37,075 tiles; $B = 2^{5^{18}}$ | | Head-filtered four-cell compiler, compared with the previous nine-cell radix $2^{5^{43}}$. Final bridge specification supersedes older comparison prose. |

| Package | Representative result | Contribution and caveat |
|---|---|---|
| 12 gap norm14 | 14/219/75; 15/44/78 | Positive auxiliary gap, shifted transport quotient, and 32-variable quartic. A shifted equality residual cannot be inserted uncharged into a unit product. |
| 13 four-cell14 | Alternative coefficient tuple | Identical local counts do not imply identical coefficients; selector-1 forbidden tiles differ from package 09. |
| 14 integer29 | 29/188/121; 29/100/130 | Only seven positive ports; canonical odd strong index and square scale; independently justified signed main-root argument. |
| 15 release | Previous round-two results | Includes 20-variable degree-eight and 25-variable sextic tradeoffs. Its final-form JSON is byte-identical to the previous delivery. |

The prior positive single-polynomial frontier includes 13/515/85, 13/571/84, 14/342/73, 14/219/75, 14/121/76, and the 15-variable alternatives of degrees $155, 123, 99, 93, 83, 56, 44$ at respectively $67, 68, 69, 70, 71, 74, 78$ operations. A frontier is relative to the delivered candidates, not a lower bound over all Diophantine constructions. The new twelve-variable row improves unknown count but not arithmetic cost or degree. The four new 15-variable rows offer further nondominated degree/cost choices.

All 62 literal upstream circuits passed independent operation, liveness and modular-line degree diagnostics. These numerical degree checks use small test coefficient values, not the materialized tuple, and do not prove a uniform fixed-coefficient degree claim. The symbolic leading-form arguments below supply that missing part for the new constructions. The upstream replay drivers have omitted vendored or ignored dependencies; they were not represented as successfully run in place.

## 3 The common normalized core

Every symbol below other than the twelve explicitly supplied coordinates and the four fixed ports denotes a computed polynomial expression. Write

$$\begin{aligned} &q = \beta J + 1, \quad C = q - F - \alpha, \quad X = wq, \quad Y = sq^3, \quad E = XY, \\ &k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H = (a+2)^2 - 1. \end{aligned} \tag{1}$$

Set

$$\begin{aligned} L &= E(kY), & A_1 &= 1 + L(L+k), \quad A_2 = 1 + \Delta c^2, \\ R &= \big((q-1)(q-F) + \alpha\big)(q^2-1) + (M_C + qM_F)J, \\ N_k &= k - hE - R, \quad B_0 = X + ac, \\ N_t &= (K_0 + w)C + q - F - z(q-1). \end{aligned} \tag{2}$$

The symbol $z$ here is the old positive transport quotient, not package 12's shifted variable. The coefficient port $M_F$ is the *shifted source port*; in Section 9 it is $M_{F,\mathrm{source}}$, not the native field mask.

The primary construction supplies

$$(J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t) \in \mathbb{Z}^{12}_{>0}.$$

The older restricted-$i$ alternative instead supplies $j > 0$ and computes $i = Hj$. The scaled-$y$ alternative keeps the same unrestricted positive $i$ as the primary construction. In either case put

$$\begin{aligned} &I = ic^2, \qquad S = \Delta I, \qquad Q = S^2, \qquad D = 1 + IS, \qquad d = Q - 1, \\ &v = cT, \qquad b = c + RD, \qquad e = Q(Dv^2 + b^2) - 1, \qquad B_a = 2Qvb. \end{aligned} \tag{3}$$

We eliminate four old roots:

$$\tau^2 = A_1, \qquad g^2 = A_2, \qquad f^2 = D, \qquad dy^2 = e - B_a f = Q(vf - b)^2 - 1. \tag{4}$$

The eliminated main quotient is $\gamma = (g - B_0)/H$. Eliminating $g$ is therefore simultaneously a quotient-integrality problem; a purely formal resultant does not solve it.

Before recovering any roots, the inherited weak transport bootstrap applies even if $N_k = N_t = -1$. It gives

$$q \geq 16, \quad C \geq 0, \quad 0 < R < E - 2, \quad k > R.$$

In particular $v, b, B_a, d$ are positive. This ordering avoids dividing by an expression whose nonvanishing has not yet been established. Strict $C > 0$ need not hold in the provisional negative-unit branch.

# 4 Explicit twelve-variable norm polynomials

## 4.1 Small separation weights

Since $\Delta > 1$ and $I > 1$, $D = 1 + \Delta I^2 < \Delta^2 I^2 = S^2$. Every conjugate of $f$ has absolute value $\sqrt{D} < S$. From (4), every real conjugate of $y$ satisfies

$$|y'|^2 = \frac{Q(vf' - b)^2 - 1}{Q - 1} < 2(Sv + b)^2.$$

Thus $Y_b = 2(Sv + b)$ is an integer strict bound. Put

$$\begin{aligned} W_b &= \begin{cases} Y_b & \text{unscaled versions,} \\ dY_b & \text{scaled-}y\text{ version,} \end{cases} \qquad P = HW_b, \quad K = P + 1, \\ M &= 6W_b, \quad u = 6P = HM, \quad Z = Ht + KB_0. \end{aligned} \tag{5}$$

Here $K$ is a separation weight, not the fixed port $K_0$. In particular $K \equiv 1 \pmod{H}$. The intended primitive coordinate is

$$t = K\gamma + M\tau + y \quad \text{or} \quad t = K\gamma + M\tau + dy.$$

Because $a + 2 < 2E$ and $c < 2kY$, one has $A_2 < 16A_1$. Also $P \geq 14$, and therefore

$$u\sqrt{A_1} > K\sqrt{A_2} + HW_b, \qquad K\sqrt{A_2} > HW_b. \tag{6}$$

Indeed $u\sqrt{A_1} - K\sqrt{A_2} > (2P - 4)\sqrt{A_1} > P$. These inexpensive weights are the ones used in the final JSON circuits. Earlier audit notes used larger sufficient weights; they are not the final circuit definition.

## 4.2 Definition without radicals or hidden quantified registers

Work in the free rank-four algebra with formal generators $r, s$ and relations $r^2 = A_1$, $s^2 = A_2$. Let $W = Z - ur - Ks$. Define

$$F_{\mathrm{p}}(r, s) = (dW^2 - H^2 e)^2 - H^4 B_a^2 D, \tag{7}$$

$$F_{\mathrm{s}}(r, s) = (W^2 - H^2 de)^2 - H^4 d^2 B_a^2 D. \tag{8}$$

For the chosen version set

$$\mathcal{R} = \prod_{\epsilon,\delta\in\{-1,1\}} F(\epsilon r, \delta s), \qquad \mathcal{P}_{12} = N_k N_t(1+\mathcal{R}) - 1. \tag{9}$$

The product is invariant under both formal sign changes and is an integer polynomial in the supplied unknowns. This definition does not require the specialized algebra to be a field, and works when the radicands or their product are squares. There is no new unknown $r$ or $s$ in (9).

For an explicit radical-free evaluator, let $(\lambda, \widetilde{e}, \widetilde{B}) = (d, e, B_a)$ for the unscaled versions and $(1, de, dB_a)$ for the scaled version. Define

$$\begin{aligned}
p_0 &= \lambda(Z^2 + u^2A_1 + K^2A_2) - H^2\widetilde{e},\\
p_1 &= 2\lambda Zu, \quad p_2 = 2\lambda ZK, \quad p_3 = 2\lambda uK, \quad \ell = H^4\widetilde{B}^2 D,\\
f_0 &= p_0^2 + A_1p_1^2 + A_2(p_2^2 + A_1p_3^2) - \ell,\\
f_1 &= 2(p_0p_1 + A_2p_2p_3), \quad f_2 = 2(p_0p_2 + A_1p_1p_3), \quad f_3 = 2(p_0p_3 + p_1p_2),\\
n_0 &= f_0^2 + A_1f_1^2 - A_2(f_2^2 + A_1f_3^2),\\
n_1 &= 2(A_2f_2f_3 - f_0f_1), \qquad \mathcal{R} = n_0^2 - A_1n_1^2.
\end{aligned}$$

These formulas are two successive quadratic norms. The archive contains the fully shared binary-gate schedules; Appendix A prints the complete primary schedule. The three-equation alternatives are simply $N_k = N_t = 1$, $\mathcal{R} = 0$.

### 4.3 The product guard

Since $B_a$ is even, $F$ is a square modulo four in the free algebra. Consequently $\mathcal{R}$ is a square modulo four, as also follows from $n_1$ being even in the explicit evaluator. If $\mathcal{P}_{12} = 0$, the three integer factors $N_k, N_t, 1+\mathcal{R}$ are units. The last cannot be $-1$, since that would require $\mathcal{R} = -2$, which is not a square modulo four. Therefore

$$\mathcal{R} = 0, \qquad N_k = N_t = \epsilon, \qquad \epsilon \in \{-1, 1\}.$$

No unproved assertion that every unit factor is positive is being used.

## 5 Why the norm does not introduce spurious roots

### 5.1 Rationality by separation of actual field conjugates

At a zero of $\mathcal{R}$, choose real square roots $\tau^2 = A_1$, $g^2 = A_2$ for a vanishing factor. In the primary case set

$$y = (Z - u\tau - Kg)/H, \qquad f = (H^2e - d(Z - u\tau - Kg)^2)/(H^2B_a).$$

In the scaled version put instead

$$\rho = (Z - u\tau - Kg)/H, \qquad f = (H^2de - (Z - u\tau - Kg)^2)/(H^2dB_a), \qquad y = \rho/d.$$

The factor identity implies $f^2 = D$ and respectively $dy^2 = e - B_af$ or $\rho^2 = d(e - B_af)$. Every displayed denominator is nonzero by the earlier bootstrap.

The field $\mathbb{Q}(\tau, g)$ is a totally real Galois extension, possibly of degree one, two or four. The derived $y$ or $\rho$ and $f$ belong to that field. Every automorphism preserves the rational integer $Z$ in

$$Z = u\tau + Kg + Hy \quad \text{or} \quad Z = u\tau + Kg + H\rho.$$

All actual conjugates of the final term have absolute value less than $W_b$. If an automorphism flipped $\tau$, subtracting the two identities would contradict the first strict inequality in (6). Thus every automorphism fixes $\tau$. The second strict inequality then prevents a flip of $g$. Both are rational and hence integral, since their squares are integers. The last term and $f$ are then rational, and $f$ is integral for the same reason.

This proof only quantifies over actual automorphisms. It does not falsely identify all independent formal sign choices with field automorphisms when the radicals are dependent. This is the point that makes the argument valid at nongeneric specializations.

### 5.2 Primary denominator recovery: saturate first

In the primary unscaled construction, the previous Galois argument has already made $\tau, g, f$ integral and $y$ rational. Put $V = vf - b \in \mathbb{Z}$. The auxiliary identity is

$$S^2V^2 - (S^2 - 1)y^2 = 1.$$

Lemma 1 below therefore gives $y \in \mathbb{Z}$ without any condition on $i$ modulo $H$. Only now recover $\gamma = (g - B_0)/H$. The identity $t - M\tau = K\gamma + y$ gives $K\gamma \in \mathbb{Z}$, and $K \equiv 1 \pmod{H}$ forces $\gamma \in \mathbb{Z}$.

This order is the final simplification: rational Pell saturation must precede quotient recovery. It removes both the computed factor $i = Hj$ and the need to merge the scaled root $dy$. Completeness keeps every old positive normalized witness and sets $t = K\gamma + M\tau + y > 0$. There is no new period, congruence or canonical divisibility choice.

### 5.3 Earlier restricted-$i$ alternative and its period repair

In the earlier restricted-$i$ alternative $i = Hj$, so $Q$ is divisible by $H$ and $\gcd(H, d) = 1$. The denominator of rational $y$ divides $H$ from its linear formula, while its square divides $d$ from $dy^2 \in \mathbb{Z}$. Hence $y \in \mathbb{Z}$. The identity

$$t - M\tau = K\gamma + y, \qquad \gamma = (g - B_0)/H$$

then gives $K\gamma \in \mathbb{Z}$. Because $\gcd(K, H) = 1$, $\gamma \in \mathbb{Z}$.

Completeness requires regenerating a normalized auxiliary witness with $H \mid i$. Merely taking the old index $m = 2cRH$ is *not* a valid argument. For instance $a = 2$, $a + 2 = 4$, $\Delta = 15$, $H = 11$, $R = 3$ and $c = \psi_4(3) = 63$ give $\psi_4(4158) = 3 \pmod{11}$, not zero.

Here is the repair. For $N = Hc^2$, the element $(a + 2) + \xi$ is a unit in

$$(\mathbb{Z}/N\mathbb{Z})[\xi]/(\xi^2 - \Delta),$$

with inverse $(a+2)-\xi$. Its finite order $e_0$ is at most $N^2$. Choose $m = 2cRe_0$, or the completely bounded proof-only choice $m = 2cR(N^2)!$. Then $N \mid \psi_{a+2}(m)$. Set

$$f = \chi_{a+2}(m), \quad i = \psi_{a+2}(m)/c^2, \quad j = i/H, \quad S = \Delta\psi_{a+2}(m), \quad V = \chi_S(R)/S, \quad y = \psi_S(R).$$

The canonical native mask supplies $R \equiv 3 \pmod{4}$. The usual odd-index quotient identities give $V \equiv -c \pmod{f}$ and $V \equiv -R \pmod{c}$. Together with $f^2 \equiv 1 \pmod{c}$ these prove that

$$T = (V + c + Rf^2)/(cf)$$

is a positive integer. The normalized strong and auxiliary equations are restored with $j > 0$, and $t = K\gamma + M\tau + y > 0$ completes the restricted-$i$ construction. The enormous period or factorial occurs only in the proof of existence, not in the polynomial or its operation ledger.

### 5.4 Rational Pell saturation and the scaled alternative

In the scaled version, rational $\rho$ has integral square $d(e-B_a f)$, so $\rho \in \mathbb{Z}$. The same congruence argument makes $\gamma$ integral. It remains to prove that $y = \rho/d$ is integral, although its denominator need not be coprime to $H$.

**Lemma 1** (Rational auxiliary Pell saturation)**.** *If $S > 1, V \in \mathbb{Z}$, $y \in \mathbb{Q}$, and $S^2V^2 - (S^2-1)y^2 = 1$, then $y \in \mathbb{Z}$.*

*Proof.* Write $S^2-1 = D_0 b_0^2$ with squarefree $D_0 > 1$. The number $b_0 y$ is integral: its reduced denominator squared would otherwise divide the squarefree integer $D_0$. Both $(S, b_0)$ and $(|SV|, |b_0 y|)$ are positive integral solutions of $x^2 - D_0 z^2 = 1$. Write them as powers $m, n$ of the least positive integral Pell unit, with coordinates $x_j, z_j$.

Since $x_m = S$ divides $x_n$, reduce $n$ modulo $2m$. In the quadratic ring modulo $x_m$, the unit to power $2m$ equals $-1$. Reflecting the remainder gives a first coordinate congruent up to sign to $x_r$, $0 \le r \le m$. If $r < m$, then $1 \le x_r < x_m$, impossible. Hence $n \equiv m \pmod{2m}$. Expanding a positive power of $x_m + z_m\sqrt{D_0}$ shows $z_m \mid z_n$. Therefore $b_0 \mid b_0 y$, and $y$ is an integer. The fundamental unit here belongs to the integer-coordinate Pell equation, not a potentially half-integral larger ring of integers. □

Apply the lemma with $V = vf - b$ and $Q = S^2$. This proves integrality without a restriction on $i$. Completeness simply keeps all old positive normalized witnesses and sets $t = K\gamma + M\tau + dy > 0$; no modular period is needed.

### 5.5 Signed recovery is an inherited theorem, not a sign convention

The recovered $f, \tau, y, g, \gamma$ may initially be signed. Replace $\tau, y$ by their absolute values. If necessary replace $f, T$ simultaneously by their negatives, preserving $f^2$ and $Tf$. The exact signed-domain extension from the pinned sources permits the resulting signs. Its strong-rank congruences give $R = \pm p \pmod{c}$, where the main lower Pell coordinate is $c = \psi_{a+2}(p)$. Bounds $0 < R, p < c/2$ force $p = R$ independently of the signs of $T, V, g$.

The first-index bound and Pell ratio then force $N_k = N_t = 1$. Before presuming $g > 0$, that ratio gives $a > X2^R$. A negative $g$ would imply $H \mid X2^R + 1$, whereas $0 < X2^R + 1 < H$. Thus $g > 0$, the projection yields $X = 2^R$, and $\gamma > 0$ follows from the Pell coordinate identities. The remaining half-binomial, population and native-window decoding does not inspect the auxiliary signs. This is an equivalence in existence, not a coordinate-preserving bijection between every pair of witness tuples.

## 6 Exact operation counts and degrees

The primary saturated product has 139 gates: 93 multiplications and 46 additions/subtractions. Its three-equation system has 135 gates: 91 multiplications and 44 additions/subtractions, with residual degrees $(7, 2, 1248)$. Every gate, supplied input and fixed port is live. The two earlier products have 140 gates, 94 multiplications and 46 additions/subtractions; their systems have 136 gates and residual degrees $(7, 2, 1536)$ or $(7, 2, 1616)$.

For the primary saturated construction,

| | $A_1$ | $A_2$ | $H$ | $I$ | $S$ | $Q$ | $D$ | $v$ | $b$ | $Y_b$ | $K$ | $u$ | $Z$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| deg | 22 | 22 | 6 | 11 | 23 | 46 | 34 | 6 | 38 | 38 | 44 | 44 | 55 |

Let $L_{\text{top}}$ denote the degree-11 leading form of $L$. Both $A_1, A_2$ have leading form $L_{\text{top}}^2$, and $B_0$ has leading form $L_{\text{top}}$. The four leading linear forms $Z - ur - Ks$ have coefficients

$$1 - 6\epsilon - \delta \in \{-6, -4, 6, 8\}$$

times $K_{\mathrm{top}}L_{\mathrm{top}}$, all nonzero. The term $H^2e$ has smaller degree than $dW^2$, and the terminal subtracted term in (7) has smaller degree than its squared bracket. Consequently

$$\mathcal{R}_{\mathrm{top}} = 1152^4 d_{\mathrm{top}}^8 (K_{\mathrm{top}}L_{\mathrm{top}})^{16}, \qquad \deg \mathcal{R} = 8 \cdot 46 + 16(44 + 11) = 1248.$$

The nonzero leading forms of $N_k, N_t$ add $7+2$, so the product degree is exactly 1257. In the restricted-$i$ alternative, $\deg d = 58$ and $\deg K = 56$ give $1536 + 9 = 1545$. In the scaled alternative the leading form is $1152^4 (K_{\mathrm{top}}L_{\mathrm{top}})^{16}$ and $\deg K = 90$ gives $1616 + 9 = 1625$. These are uniform symbolic arguments for authentic $\beta > 0$; finite-field specializations are diagnostics only.

## 7 A polynomial in 26 unrestricted integers

The smaller integer lift uses the primary unscaled norm and saturation before quotient recovery. Retain positivity only for

$$J, F, \alpha, z, s, \eta, \zeta,$$

and take $h, T, t$ unrestricted, $i = 2r_i + 1$ odd, and $w = r_w^2$. Thus $i \neq 0$, while $w$ is provisionally allowed to vanish. The polynomial bound valid for signed $i, T$ is

$$W_b = 2\big((Si)v^2 + b\big), \quad P = HW_b, \quad K = P + 1, \quad M = 6W_b, \quad u = 6P. \tag{10}$$

Here $Si = \Delta i^2 c^2 \geq |S|$ and $v^2 \geq |v|$ for integral $v$, so $W_b \geq 2(|Sv| + b)$. No absolute-value gate occurs in the circuit. Use $F_{\mathrm{p}}$ from (7), not the scaled norm.

If $B_a \neq 0$, root recovery gives $f^2 = D$ and $dy^2 = e - B_a f$. If $B_a = 0$, then $T = 0$ and the factor equation directly gives $dy^2 = e = Qb^2 - 1$. In either case all actual conjugates satisfy $|y'| < W_b$. No division by $B_a$ is performed on its zero branch.

If $w = 0$, then $X = E = L = 0$ and $A_1 = 1$, so the first recovered root is integral. The second separation inequality makes $g$ integral and $y$ rational. When $B_a \neq 0$, the recovered $f$ is a rational integer and Lemma 1, applied to $V = vf - b$, makes $y$ integral. When $B_a = 0$, apply the same lemma directly to the integral $V = -b$, without needing $f$. Now $K\gamma = t - M\tau - y \in \mathbb{Z}$ and $\gcd(K, H) = 1$ give $\gamma \in \mathbb{Z}$. The main norm with $g = ac + \gamma H$ reduces to $1 = 0 \pmod{H}$, impossible. This excludes $w = 0$ before using the positive-scale first-root estimate.

For $w \geq 1$, $X \geq q \geq 16$ and full Galois separation makes $\tau, g$ integral and $y$ rational. The same two saturation branches make $y$ integral and then $\gamma$ integral. The weak transport bounds imply $h \geq 0$, $0 < R < a < \Delta$ and $1 < b < Q$. If $T = 0$, the already established equation $Qb^2 - (Q-1)y^2 = 1$ gives

$$(Q-1)(y^2 - b^2) = b^2 - 1 > 0,$$

forcing $|y| \geq b + 1$ and contradicting $1 < b < Q$. Thus $B_a \neq 0$, all divisions are justified, and the complete signed normalized source tuple is restored. Canonical completeness witnesses have odd $i$, square $w$, and $t = K\gamma + M\tau + y$; no new divisibility condition is required.

Use the standard surjective positivity map

$$\mathrm{Pos}(a, b, c) = 1 + a^2 + b^2 + c(c+1) : \mathbb{Z}^3 \longrightarrow \mathbb{Z}_{>0}$$

for each of the seven positive ports. Its surjectivity is the classical two-squares-plus-pronic representation, obtained from the three-square theorem applied to $4n - 3$ for a target integer $n \geq 1$. Together with $h, T, t, r_i, r_w$, this gives $7 \cdot 3 + 5 = 26$ unrestricted integers.

The circuit has 192 gates: 117 multiplications and 75 additions/subtractions. Giving weight two to the seven positive ports and $w$, and weight one to $i, h, T, t$, one has $\deg d = 90$, $\deg K = 86$ and

$\deg L = 22$. Thus the resultant degree is $8 \cdot 90 + 16(86 + 22) = 2448$, and the outer units add $13 + 4$, giving exactly 2465. A uniform nonzero monomial certificate after all substitutions is

$$[J_a^{1578} h r_i^{48} \eta_a^{224} r_w^{194} s_a^{418} z_a^2]\, \mathcal{P}_{26} = 2^{124} 3^8 \beta^{789} \neq 0.$$

Its exponent sum is 2465. The earlier scaled-$y$ 26-variable construction, degree 3185 at 194 operations, remains in the archive as a fully proved but dominated alternative. The two 28-integer constructions, degrees 426 and 258 at 127 and 128 gates, remain useful lower-degree tradeoffs.

### 7.1 A 27-integer intermediate tradeoff

The independently audited weighted three-root merger retains $\gamma$ and eliminates $f, \tau, y$. Use the same seven positive ports and odd-$i$, square-$w$ substitutions, with unrestricted $h, T, \gamma, t, r_i, r_w$. Set

$$\begin{aligned} M_3 &= D + v^2, \qquad C_3 = d(t^2 + M_3^2 A_1) - e, \\ U_3 &= C_3^2 + D B_a^2 - 4 d^2 t^2 M_3^2 A_1, \\ \mathcal{R}_3 &= U_3^2 - D(2 C_3 B_a)^2, \qquad \mathcal{P}_{27} = N_1 N_k N_t (1 + \mathcal{R}_3) - 1, \end{aligned}$$

where $N_1 = (B_0 + \gamma H)^2 - \Delta c^2$. The main norm excludes $w = 0$. With signed $h$, the correct bootstrap is $h \geq 0$, $R \leq k + 1 < 2k$, not $h > 0$ or $k > R$. It yields $A_1 > 16(c^2 + R^2) + 1$, hence

$$d A_1 M_3^2 > 2Q(D v^2 + b^2) = 2(e + 1), \qquad C_3 > e + 2.$$

This excludes $t = 0$ before any division by it. The $T = 0$ case directly restores integral $\tau, y$ and contradicts $1 < b < Q$, so $B_a \neq 0$. Now

$$f = -U_3/(2 C_3 B_a), \quad \tau = (C_3 + B_a f)/(2 d t M_3), \quad y = t - M_3 \tau$$

are rational roots whose integer squares force their integrality. The inherited signed theorem supplies soundness, and canonical witnesses with $t = M_3 \tau + y$ supply completeness. Seven positivity maps plus six free integers give 27 variables. The exact cost is 136 operations, 72 multiplications and 64 additions/subtractions, and the exact degree is 1116. A nonzero top coefficient is

$$[J_a^{688} h \gamma r_i^{24} \eta_a^{114} r_w^{86} s_a^{200} z_a^2] \mathcal{P}_{27} = 2^{27} \beta^{344}.$$

The full sign and denominator proof, 150 independent formula checks and two modular coefficient extractions are included with this middle tradeoff.

## 8 Fewer operations at degree 44

For the ordinary fifteen-variable core, keep $g = B_0 + \gamma H$ supplied through $\gamma$ and write

$$\begin{aligned} N_0 &= \tau^2 - L(L + k), & N_1 &= g^2 - \Delta c^2, \\ U &= \Delta(f^2 - 1), & V &= c(Tf - 1) - R f^2, \\ N_a &= U(V^2 - y^2) + y^2, & S_o &= i^2 c^4 - U. \end{aligned}$$

Use the same $N_k, N_t$ as before. A first gap replaces $\tau$ by $L + t_0$, and an auxiliary gap replaces $y$ by $V + e_0$, with $t_0, e_0 > 0$. The exact useful identities are

$$N_0 = t_0^2 + L(2t_0 - k), \qquad N_a = (V + e_0)^2 - U e_0 (2V + e_0).$$

The following four literal polynomials use the indicated gaps:

$$\begin{aligned}
Q_{44} &= (N_0N_k-1)^2+(N_1N_t-1)^2+(N_a-1)^2+S_o^2 && \text{both},\\
Q_{50} &= (N_0N_t-1)^2+(N_1N_k-1)^2+(N_a-1)^2+S_o^2 && \text{auxiliary},\\
Q_{70} &= (N_0N_1N_t-1)^2+(N_aN_k-1)^2+S_o^2 && \text{first},\\
Q_{80} &= (N_0N_1-1)^2+(N_aN_kN_t-1)^2+S_o^2 && \text{neither}.
\end{aligned}$$

A zero forces all five unit expressions to be integer units, with product one and $S_o = 0$. Hence it gives a zero of the inherited ordinary product. The first and main norm and auxiliary norm exclude negative units by their Pell/modulo-four arguments. For $Q_{80}$ the linear units remain grouped together, so its soundness uses the full inherited product sign recovery, not an assertion of their automatic positivity.

The restored $y = V + e_0$ need not be positive off the zero set. On that set it is nonzero; replacing it by $|y|$ leaves the source equations unchanged. Completeness uses the canonical positive witnesses, for which $\tau > L$ and $y > V$. Thus these are valid existence-preserving gap transformations.

The ordinary shared expressions cost 61 operations; each gap adds one. Partitioning the five units into $r$ groups, forming residuals, squaring and summing adds $7 + 2r$ operations. This gives costs $76, 75, 73, 72$. Each has 40 multiplications and respectively $36, 35, 33, 32$ additions/subtractions. The residual degree lists are respectively

$$(19, 20, 22, 22), \quad (24, 25, 22, 22), \quad (32, 35, 22), \quad (40, 37, 22).$$

Their highest real sums of squares cannot cancel, giving exact total degrees $44, 50, 70, 80$. In particular the degree-44 construction improves the pinned 78-gate schedule by two operations without adding unknowns.

# 9 A 943-tile compiler with materialized coefficients

## 9.1 The literal machine and directed-signal tiles

Use the exact 430-action K215 table at the reviewed pin. Its SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

The initial state is 0 on a blank two-sided binary tape; states 0–214 work and 215 halts. The first action is $(0, 0) \mapsto (1, \mathrm{R}, 93)$. The original write and movement on every transition entering halt are retained. Only after entry add two absorbing stay loops.

A tile lists $(\text{west}, \text{north}, \text{east}, \text{south})$. Vertical colors are headless bits 0,1, head codes $\mathrm{head}(q, b) = 2(q + 1) + b$, and $L = 434, Q = 435, R = 436, V = 437, H_b = 438$. Horizontal colors are $O = 0, HB = 1, PL = 2, PQ = 3, PR = 4, BAD = 5$, followed by directed signals. Let $S_R, S_L$ be the sorted target-state sets of original right- and left-moving actions; their sizes are 199 and 45. Define $\mathrm{right}(q) = 6 + \mathrm{index}_{S_R}(q)$ and $\mathrm{left}(q) = 205 + \mathrm{index}_{S_L}(q)$.

The complete finite tile recipe is:

1. For every working right action $(q, b) \mapsto (v, \mathrm{R}, t)$ take $(O, \mathrm{head}(q, b), \mathrm{right}(t), v)$; for a left action take $(\mathrm{left}(t), \mathrm{head}(q, b), O, v)$: 430 tiles.
2. Two halt loops $(O, \mathrm{head}(215, b), O, \mathrm{head}(215, b))$ and two headless stays $(O, b, O, b)$.
3. Arrival tiles $(\mathrm{right}(q), b, O, \mathrm{head}(q, b))$ and $(O, b, \mathrm{left}(q), \mathrm{head}(q, b))$: $398 + 90 = 488$ tiles.
4. Four finish tiles $(O, c, O, H_b)$ for $c \in \{0, 1, \mathrm{head}(215, 0), \mathrm{head}(215, 1)\}$.
5. Four initialization tiles $(PL, L, PL, 0)$, $(PL, Q, PQ, 1)$, $(PQ, R, PR, \mathrm{head}(93, 0))$, $(PR, R, PR, 0)$.

6. Three reset tiles $(HB, H_b, HB, p)$ for $p \in \{L, Q, R\}$.

7. Nine seam tiles $(u, V, v, V)$, with $u \in \{O, HB, PR\}$ and $v \in \{O, HB, PL\}$ independently.

8. One forbidden tile $(BAD, 0, O, 0)$.

The total is $430 + 2 + 2 + 488 + 4 + 4 + 3 + 9 + 1 = 943$, with no duplicates. Start is $(PL, Q, PQ, 1)$, selector 0; Forbidden is selector 1; the rest are lexicographically ordered. BAD never occurs as an east color, so Forbidden is globally impossible. The JSON contains the exact ordered finite list.

### 9.2 Global marked-tiling equivalence, including seams

V occurs on one vertical side exactly when it occurs on both, so V columns persist in time. From a Start, the horizontal phase chain must meet V columns on both sides by periodicity; there is no phase cycle containing Q. Between the nearest V columns the initialization row is $L^u Q R^v$ with $u \geq 0, v \geq 1$. Its predecessor is an H-reset row. The initialization implements the literal first action, and directed signals thereafter transfer the unique head exactly one cell in its genuine direction. A head cannot cross V, since no V seam accepts a directed state signal.

Temporal periodicity supplies a subsequent reset row. Immediately above its first occurrence every cell must be a finish tile, and the unique head is halted. A finish cannot mix with an ordinary successor across that strip: in the next row HB edges propagate across all H cells, while ordinary tiles have no HB edge. This proves soundness without assuming a globally unique Start.

Conversely, pad a finite blank halting run with at least two never-visited blank cells next to each V boundary, and with absorbing halt rows. Width $w_c$ and height $h_c$ may be any sufficiently large integers, independently, hence powers of five. At the V seam, choose its west color from time $t - 1$ on the last cell and its east color from time $t$ on the first cell. The nine independent seam choices are exactly what is needed for

$$\mathrm{tile}(x, t) = G(x \bmod w_c, (t + \lfloor x/w_c \rfloor) \bmod h_c).$$

This has periods $(w_c, -1)$ and $(0, h_c)$. The cyclic coordinate $i = -(x - x_{\mathrm{Start}}) - w_c(t - 1) \pmod{w_c h_c}$ puts Start at zero, the right neighbor at shift $-1$ and the temporal neighbor at shift $-w_c$. Conversely any cyclic word with these two matching shifts pulls back to a periodic plane, without requiring a divisibility or coprimality condition between the shifts and word length. Start's distinct north and south colors exclude temporal shift zero. This preserves all independent padding required by the native dummy-control theorem.

### 9.3 Bounded definitions of all four actual integers

All quantities in this subsection are fixed integers, not supplied unknowns. Set

$$A = 1024, \quad b_c = 625, \quad \rho_c = 2^{625}, \quad L_c = 15625, \quad d_c = b_c L_c = 9{,}765{,}625 = 5^{10}, \quad B = \rho_c^{L_c}.$$

Each tile label $x < 512$ is represented by the ten bits of $x + 512$, including an occupancy bit. In west, north, east, south order put

$$e_0(j) = 1023 + j, \ e_1(j) = 1013 + j, \ e_2(j) = 1003 + j, \ e_3(j) = 983 + j \quad (0 \leq j \leq 9).$$

The native exponent set $\mathcal{E}$ consists of selectors $0,\dots,942$, those forty copy positions, dummies 1063,1064, and anchors $1095, 1096, 2195, 3296$. Its size is 989. For tile $i$ with slot labels $x_{it}$ define

$$c_i = 1 + \sum_{t=0}^{3}\sum_{j=0}^{9} \mathrm{bit}_j(x_{it}+512)A^{1+10t+j} + \sum_{j=41}^{44} A^j,$$
$$c_{e_t(j)} = A^{1+10t+j}, \qquad c_{1095} = A^{41},\ c_{1096} = A^{42},\ c_{2195} = A^{43},\ c_{3296} = A^{44},$$
$$c_{1063} = c_{1064} = 0, \qquad \mu = A - 2 + \sum_{j=1}^{44}(A-3)A^j + 2^{450}(2^{78}-1).$$

Then the bounded finite sums defining the coefficients are

$$D_C = \rho_c^{30} + \rho_c^{5517} + \rho_c^{1100} + \rho_c^{2200} + \sum_{e\in\mathcal{E}} c_e(\rho_c^{7631-e} + \rho_c^{8733-e}),$$
$$D_R = \rho_c^{5497},$$
$$M_{F0} = \mu(\rho_c^{7631} + \rho_c^{8733}) + \sum_{j=0}^{9}\rho_c^{1013+j} + \sum_{j=0}^{9}\rho_c^{5497+1023+j} + \rho_c^{2195} + \rho_c^{3296},$$
$$M_C = B - 1 - \sum_{\substack{e\in\mathcal{E}\\ e\neq 1}} \rho_c^e - 2\rho_c^{1063}, \qquad M_{F,\mathrm{native}} = M_{F0} + 4,$$
$$\beta = B-1, \quad K_0 = D_C + BD_R, \quad M_C, \quad M_F = M_{F,\mathrm{source}} = M_{F,\mathrm{native}} + B - 1. \tag{11}$$

No recursive search or existential choice remains in these definitions. The machine table, sorted finite tile list, integer powers and bounded sums determine every byte. All four ports must transfer together; in particular binding the source $M_F$ to the unshifted native mask is wrong.

### 9.4 Why the compact masks preserve the compiler

A clause sum $\sum c_e d_e$ has no carries in base $A$: its units digit is at most 943 and every other digit at most 944, below 1024. The low mask $A-2$ allows selector count zero or one. For each bit the tested digit is selected-bit plus copy-bit; $A-3$ allows precisely zero or two. Thus empty cells have zero copies and anchors, while an occupied cell's copies equal all forty tile bits and all anchors are one. The 78 padding bits start at bit 450, above every possible clause sum, so they test identically zero data. Saturating the masks therefore adds no constraint on valid cells.

The occupancy bit propagates from Start through the whole cyclic word along the spatial shift one. The remaining bits enforce the exact directed edge equalities. This replaces the previous one-hot occupancy argument rather than simply deleting it.

The population identities are

$$\mathrm{popcount}(\mu) = 483, \qquad \mathrm{popcount}(M_{F,\mathrm{native}}) = 2\cdot 483 + 20 + 2 + 1 = 989, \qquad \mathrm{popcount}(M_C) = d_c - 989.$$

Moreover $0 < M_C, M_{F,\mathrm{native}} < B-1$, $M_C \equiv 2 \pmod 4$, and $M_{F,\mathrm{native}} \equiv 4 \pmod 8$. Hence the inherited inverse-population and exact two-bit-surplus arguments retain their hypotheses. The low-origin field test is unchanged: all new dense bits are far above it.

The compact layout uses $H_c = 5497$, $T_1 = 7631$, $T_2 = 8733$ and the exact support

$$\mathcal{E} = \big([0,942]\cup[983,992]\cup[1003,1032]\cup\{1063,1064,1095,1096\}\big)\cup\{2195,3296\}.$$

The possible upper-dummy spill position 1064 already belongs to this set. The earlier sufficient interval conditions, such as a band gap exceeding twice the support span, are no longer assumed. What is required and checked is isolation at the positions actually tested.

At $1013 + j$, bare shift 30 sees only the current south copy $983 + j$, paired with the temporal north copy. At $6520 + j$, bare shift 5517 sees east copy $1003 + j$, paired with the spatial west copy $1023 + j$ shifted by $D_R$. At anchor targets 2195 and 3296, bare shifts 1100 and 2200 see respectively 1095 and 1096. Every other bare or spatial contribution is absent. Clause contamination at a tested target would require $T_1 - t$ or $T_2 - t$ in $\mathcal{E} - \mathcal{E}$. Only the spatial targets need a nontrivial check: the resulting intervals $[1102, 1111]$ and $[2204, 2213]$ contain no such difference. The small support component has span 1096, and subtracting those intervals from the two high anchors lands in gaps of the support.

At the clause targets themselves, the bare and spatial shifts see no support. The band gap 1102 is absent from $\mathcal{E} - \mathcal{E}$: $2195 - 1102 = 1093$ and $3296 - 1102 = 2194$ are absent, and the small component is too short. Thus the two possible rotation-hit sets $(T_1 - \mathcal{E}) \bmod L_c$ and $(T_2 - \mathcal{E}) \bmod L_c$ are disjoint. Every binary rotation leaves a clean clause band. Supports may overlap away from tested positions; the uniform carry bound already includes both bands and those unused positions are not decoded.

For a nonaligned binary rotation $b_c t + \varepsilon$, $1 \le \varepsilon \le b_c - 2$ makes all rotated digits even, and $\varepsilon = b_c - 1$ permits at most one odd upper-dummy spill. Start needs odd hits at two distinct anchors, excluding both cases. For an aligned rotation the unique support pair at difference $3296 - 2195 = 1101$ is $(2195, 3296)$. There is no circular wrap because $L_c = 15625 > 2 \cdot 3296$. Thus the simultaneous anchor intersection is exactly $\{0\}$ and whole-cell alignment follows.

The no-carry threshold still has 529 bits, below $\rho_c = 2^{625}$. The largest unshifted exponent is $8733 + 3296 = 12029 < 15625$, and all reflected exponents are positive. These exact inequalities replace the old conservative spacing. Finally $\rho_c \equiv 2 \pmod 5$, the band gap is 2 (mod 4), and paired clause terms cancel, leaving

$$D_C \equiv 3 \pmod 5, \qquad D_R \equiv 2 \pmod 5, \qquad 2D_C - D_R \equiv 4 \pmod 5.$$

This is a unit, sufficient for the retained dummy-control theorem to tune the *actual packed rank*. The temporal stride is not treated as an independent exponent. Arbitrarily large power-of-five padding supplies the theorem's no-wrap hypotheses. These are the explicit interface reasons that the Wang tuple can replace the old authentic tuple.

### 9.5 Actual payloads, sizes and hashes

The four compressed payloads in `compiler/coefficients_compact/` inflate to minimal-byte unsigned *little-endian* integers with no header. Byte $j$ contributes $256^j$ times its value. The high unused bits of the last byte are zero. They are not decimal strings or big-endian network integers.

| Port | Bits | Raw bytes | Gzip bytes |
|---|---:|---:|---:|
| $\beta$ | 9,765,625 | 1,220,704 | 1,219 |
| $K_0$ | 13,201,251 | 1,650,157 | 16,111 |
| $M_C$ | 9,765,625 | 1,220,704 | 1,616 |
| $M_{F,\text{source}}$ | 9,765,626 | 1,220,704 | 1,314 |
| Total | | 5,312,269 | 20,260 |

The uncompressed SHA-256 values are:

$\beta$ `40cc309fcb5dd2942c371d18be6d955d472e6b4c3604034b316194037ac16a5d`

$K_0$ `e79f794af9565a7d824beb423bd03ee9ac117811b9b0e3a83ffe18f8a084de99`

$M_C$ `2e3107193f3b376fc3cfffdaa3ef547e6dd8dfdd4548519c0e33b5ffe2f2371d`

$M_{F,\text{source}}$
`8851b54fffd93780996de5f8c69315448e1369c3eea142428d2225a5149d1825`

The bit length of $\beta$ is 390,625 times smaller than the T4 value $5^{18}$. Gzip's small size reflects long sparse or constant runs; it is not a small-coefficient claim in the usual height sense. Literal use of these integers still involves millions of bits.

## 10 Evidence, reproducibility, and limitations

The archive separates four levels of evidence.

1. **Imported theorems.** The fixed K215 blank-halting/refutation calibration, Pell classification and congruences, native-mask and dummy-control arithmetic, and the signed-domain source extension. They are pinned and included or linked as inert source evidence. The new checkers do not re-prove them.

2. **New unbounded mathematics.** Galois separation, denominator recovery, rational Pell saturation, modular-period completeness, signed zero-branch exclusions, grouped-unit transformations, the Wang marked-plane proof and compiler interface checks. Their detailed prose is included as proof documents as well as the article.

3. **Finite checks.** Literal DAG counts and full liveness; independent quotient-algebra evaluations; modular degree diagnostics; exact tile reconstruction; geometry, populations, carry bounds, five-adic unit and byte-for-byte coefficient reconstruction.

4. **Not supplied.** An end-to-end kernel certificate, a proof that K215 does or does not halt, an astronomical accepting Pell witness, or a global optimality proof.

The positive-root checker compares 100 assignments for each variant to an independently written four-dimensional quotient-algebra norm, checks both degree diagnostics at two primes, and checks 21 synthetic radical-kernel witnesses per variant. Its 49,000 rational-Pell cases supplement Lemma 1, not replace it. The integer-26 checker performs 120 independent four-factor norm evaluations, checks a finite rational-Pell grid and verifies the explicit top coefficient at two primes. Grouped circuits are checked against separate formulas on 100 signed assignments each.

The semantic tile checker reconstructs all 943 tiles without importing the builder. Its 425 padded helices over 17 finite halting-control machines test 33,800 tiles and both edge matchings. These are seam controls, not K215 halting experiments. An independent compiler checker reconstructs the full tuple and compares all 5,312,269 binary bytes. The gzip round trips, low congruences, population counts and lengths are checked exactly.

The portable top-level checker uses Python's standard library, inflates and verifies the four payloads, audits the final DAGs and source hashes, and runs the newly authored independent compiler and integer checks. Optional original research checkers that use NumPy are clearly marked; no dependency is installed. The required SHA-256 manifest covers every packaged deliverable except the manifest itself. The public runner rejects optimized Python, copies the release into a temporary directory for mutable checks, and verifies that the original files remain unchanged, allowing repeated runs. Rebuilding the PDF uses an existing TeX installation; the build script neither downloads packages nor enables shell escape.

## 11 Further research questions and unsuccessful directions

1. **Reduce norm cost, not just its bound.** The twelve-variable improvement raises the cost from the previous 13-variable 84/85-gate frontier to 139. A more economical norm evaluator, or a different denominator-clearing primitive element, could improve this tradeoff. Any proposed cancellation must be checked at dependent-radical and zero-cross-term specializations.

2. **Remove another supplied parameter safely.** The remaining positive geometry, transport and strong-index parameters cannot be dropped merely because they have canonical values in completeness. Soundness must rule out every new zero branch. No eleven-positive-variable construction is established here.

3. **Shrink the integer lift.** The 26-integer count uses seven independent three-integer positivity maps. A reduction would need a genuinely weaker signed domain or a jointly justified positivity representation, not an uncharged change of domain. The $w = 0$ and $T = 0$ branches show why the issue is substantive.

4. **Go below 67 operations.** Grouping improves low-degree schedules but does not beat the existing minimum-operation polynomial. The 60-operation comparison system does not immediately imply a 66-operation product because its shifted residual is not already the necessary unit.

5. **Improve the quartic independently.** Bounded direct-elimination searches from the pinned 32-variable quartic, including positive root shears, did not establish fewer than 32 positive variables at degree four. The bounded beam search is not a lower bound. The 25-variable sextic and 20-variable degree-eight forms remain different tradeoffs.

6. **Reduce coefficient height further.** Wang943 makes materialization practical. Tighter signal alphabets, binary encodings or anchor geometry may reduce it again, but local tile counts alone do not prove a global compiler. Preserve the marked periodic-plane argument, helical padding, both population identities and actual-rank dummy tuning.

7. **Formalize the bridge.** The largest remaining verification gap is a kernel-checked composition of machine calibration, compiler, signed Pell restoration and the new radical elimination. Finite byte checks and independently reviewed prose do not close it by themselves.

## 12 Pinned references and archive map

The repository root for the reviewed family is

https://github.com/VladimirReshetnikov/ProveIt/tree/f0ae4e0991ca44a7bb4e5671d42e33d775e48035/SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/con-zfc-diophantine-forms.

The exact K215 table and endpoint are under `Computability/BusyBeaver/Research/ZFC/candidate-k215/k215.tm` and `Computability/BusyBeaver/Lean/BusyBeaver/ZFC/KEndpoint215.lean` at the same pin. Full immutable per-file URLs and hashes are in `review/SOURCE_PINS.json`; supplementary source manifests identify inherited signed-domain and normalized-core dependencies.

- `report/`: this TeX source, PDF, printable primary circuit and build script.
- `roots/`: six literal twelve-variable circuit JSONs, mathematical proof and check receipts.
- `integers/`: 26-, 27- and 28-integer circuits, signed-domain proofs and independent checks.
- `arithmetic/`: the four grouped schedules, formulas and audit receipt.
- `compiler/`: exact tile list, scalar layout, full bounded recipe, deterministic coefficient payloads and newly authored generators.
- `audit/`: independent mathematical reviews and finite coefficient reconstruction.
- `review/`: inert pinned source records, comparison and the 62-circuit audit.

- `COEFFICIENT_BINDING.json`, `FINAL_FORMS.json`, `verify_package.py` and `SHA256SUMS.txt`: release-level binding, metrics and integrity checks.

The release-level binding is authoritative for the new Wang943 instance; source JSON provenance may retain its original coherent historical tuple. This is an explicit all-four-port rebinding, not a silent edit to inherited data.

# A Complete primary binary-gate schedule

The following 139 rows are generated verbatim from `root12_saturated_product.json`. Inputs are the twelve coordinates listed in Section 3 and fixed ports `beta,K0,MC,MF`; every other name is assigned exactly once. A row reads destination = left operand, operator, right operand. This is a complete integer-polynomial definition with sharing. The alternate scaled and integer schedules are included as exact JSON rather than silently abbreviated formulas.

| Gate | Destination | Left | Op | Right |
|---|---|---|---|---|
| 1 | rep | beta | * | J |
| 2 | q | rep | + | 1 |
| 3 | q2 | q | * | q |
| 4 | q3 | q2 | * | q |
| 5 | X | w | * | q |
| 6 | Y | s | * | q3 |
| 7 | E | X | * | Y |
| 8 | k | eta | + | zeta |
| 9 | kY | k | * | Y |
| 10 | Lfirst | E | * | kY |
| 11 | first_next | Lfirst | + | k |
| 12 | first_product | Lfirst | * | first_next |
| 13 | c | kY | + | eta |
| 14 | a | E | + | Y |
| 15 | ac | a | * | c |
| 16 | Dbase | X | + | ac |
| 17 | a4 | 4 | * | a |
| 18 | H | a4 | + | 3 |
| 19 | a2 | a | * | a |
| 20 | Delta | a2 | + | H |
| 21 | c2 | c | * | c |
| 22 | Dc2 | Delta | * | c2 |
| 23 | qF | q | − | F |
| 24 | C | qF | − | alpha |
| 25 | hE | h | * | E |
| 26 | index_part | k | − | hE |
| 27 | gap_product | rep | * | qF |
| 28 | G | gap_product | + | alpha |
| 29 | q2m1 | q2 | − | 1 |
| 30 | Rprod | G | * | q2m1 |
| 31 | qMF | q | * | MF |
| 32 | mask_factor | MC | + | qMF |
| 33 | mask | mask_factor | * | J |
| 34 | R | Rprod | + | mask |
| 35 | Nk | index_part | − | R |
| 36 | Kplusw | K0 | + | w |
| 37 | innerC | Kplusw | * | C |
| 38 | trans_part | innerC | + | qF |
| 39 | zrep | z | * | rep |
| 40 | Nt | trans_part | − | zrep |
| 41 | Afirst | first_product | + | 1 |
| 42 | Gmain | Dc2 | + | 1 |
| 43 | Istrong | i | * | c2 |
| 44 | Saux | Delta | * | Istrong |
| 45 | Qaux | Saux | * | Saux |
| 46 | Dstrongm1 | Istrong | * | Saux |
| 47 | Dstrong | Dstrongm1 | + | 1 |
| 48 | daux | Qaux | − | 1 |
| 49 | vaux | c | * | T |

| Gate | Destination | Left | Op | Right |
|---|---|---|---|---|
| 50 | r12_50 | vaux | * | vaux |
| 51 | r12_51 | R | * | Dstrong |
| 52 | baux | c | + | r12_51 |
| 53 | r12_53 | Dstrong | * | r12_50 |
| 54 | r12_54 | baux | * | baux |
| 55 | Dv2plusb2 | r12_53 | + | r12_54 |
| 56 | r12_56 | Qaux | * | Dv2plusb2 |
| 57 | eaux | r12_56 | − | 1 |
| 58 | r12_58 | vaux | * | baux |
| 59 | r12_59 | Qaux | * | r12_58 |
| 60 | Baux | 2 | * | r12_59 |
| 61 | r12_61 | Saux | * | vaux |
| 62 | r12_62 | r12_61 | + | baux |
| 63 | Ybound | 2 | * | r12_62 |
| 64 | Pweight | H | * | Ybound |
| 65 | Kweight | Pweight | + | 1 |
| 66 | r12_66 | H | * | t |
| 67 | r12_67 | Kweight | * | Dbase |
| 68 | Zlinear | r12_66 | + | r12_67 |
| 69 | uweight | 6 | * | Pweight |
| 70 | H2 | H | * | H |
| 71 | r12_71 | Zlinear | * | Zlinear |
| 72 | r12_72 | uweight | * | uweight |
| 73 | r12_73 | r12_72 | * | Afirst |
| 74 | r12_74 | r12_71 | + | r12_73 |
| 75 | r12_75 | Kweight | * | Kweight |
| 76 | r12_76 | r12_75 | * | Gmain |
| 77 | r12_77 | r12_74 | + | r12_76 |
| 78 | r12_78 | daux | * | r12_77 |
| 79 | r12_79 | H2 | * | eaux |
| 80 | P00 | r12_78 | − | r12_79 |
| 81 | r12_81 | 2 | * | daux |
| 82 | r12_82 | Zlinear | * | uweight |
| 83 | P10abs | r12_81 | * | r12_82 |
| 84 | r12_84 | Zlinear | * | Kweight |
| 85 | P01abs | r12_81 | * | r12_84 |
| 86 | r12_86 | uweight | * | Kweight |
| 87 | P11 | r12_81 | * | r12_86 |
| 88 | r12_88 | H2 | * | H2 |
| 89 | r12_89 | Baux | * | Baux |
| 90 | r12_90 | r12_89 | * | Dstrong |
| 91 | Lnorm | r12_88 | * | r12_90 |
| 92 | r12_92 | P00 | * | P00 |
| 93 | r12_93 | P10abs | * | P10abs |
| 94 | r12_94 | Afirst | * | r12_93 |
| 95 | r12_95 | r12_92 | + | r12_94 |
| 96 | r12_96 | P01abs | * | P01abs |
| 97 | r12_97 | P11 | * | P11 |
| 98 | r12_98 | Afirst | * | r12_97 |
| 99 | r12_99 | r12_96 | + | r12_98 |
| 100 | r12_100 | Gmain | * | r12_99 |
| 101 | r12_101 | r12_95 | + | r12_100 |
| 102 | F00 | r12_101 | − | Lnorm |
| 103 | r12_103 | P00 | * | P10abs |
| 104 | r12_104 | P01abs | * | P11 |
| 105 | r12_105 | Gmain | * | r12_104 |
| 106 | r12_106 | r12_103 | + | r12_105 |
| 107 | F10abs | 2 | * | r12_106 |
| 108 | r12_108 | P00 | * | P01abs |
| 109 | r12_109 | P10abs | * | P11 |
| 110 | r12_110 | Afirst | * | r12_109 |
| 111 | r12_111 | r12_108 | + | r12_110 |
| 112 | F01abs | 2 | * | r12_111 |
| 113 | r12_113 | P00 | * | P11 |
| 114 | r12_114 | P10abs | * | P01abs |
| 115 | r12_115 | r12_113 | + | r12_114 |
| 116 | F11 | 2 | * | r12_115 |
| 117 | r12_117 | F00 | * | F00 |
| 118 | r12_118 | F10abs | * | F10abs |

| Gate | Destination | Left | Op | Right |
|---|---|---|---|---|
| 119 | r12_119 | Afirst | * | r12_118 |
| 120 | r12_120 | r12_117 | + | r12_119 |
| 121 | r12_121 | F01abs | * | F01abs |
| 122 | r12_122 | F11 | * | F11 |
| 123 | r12_123 | Afirst | * | r12_122 |
| 124 | r12_124 | r12_121 | + | r12_123 |
| 125 | r12_125 | Gmain | * | r12_124 |
| 126 | N00 | r12_120 | − | r12_125 |
| 127 | r12_127 | F01abs | * | F11 |
| 128 | r12_128 | Gmain | * | r12_127 |
| 129 | r12_129 | F00 | * | F10abs |
| 130 | r12_130 | r12_128 | − | r12_129 |
| 131 | N10 | 2 | * | r12_130 |
| 132 | r12_132 | N00 | * | N00 |
| 133 | r12_133 | N10 | * | N10 |
| 134 | r12_134 | Afirst | * | r12_133 |
| 135 | Res12 | r12_132 | − | r12_134 |
| 136 | unit12 | Res12 | + | 1 |
| 137 | r12_137 | Nk | * | Nt |
| 138 | r12_138 | r12_137 | * | unit12 |
| 139 | polynomial12 | r12_138 | − | 1 |